

Reply to: “Association between antiviral treatments and fracture in elderly patients with HBV needs further evaluation”

To the Editor:

We would like to express our gratitude to Dr. Wu and colleagues¹ for their insightful comments on our study.² Our study was a territory-wide cohort study recruiting patients captured by the Clinical Data Analysis and Reporting System (CDARS) of the Hospital Authority, Hong Kong. Patients were from all tiers of medical services – from primary care clinics to tertiary hospitals – and this provided a wide spectrum of patients with various health statuses and comorbidities to our study. Acknowledging the lack of data on bone mineral density in our study, we adopted the diagnosis codes for osteoporosis and history of fracture, as well as established risk factors of osteoporosis, such as smoking and drinking statuses, rheumatoid arthritis and endocrinopathies to identify those with underlying osteoporosis or at risk of osteoporosis.³ The use of diagnosis codes from CDARS was highly specific and accurate with prior internal validation.⁴ Moreover, the use of steroids, calcium and vitamin D supplements, and anti-resorptive therapies were captured at the same time. These parameters were included to estimate the propensity score for subsequent matching between the entecavir (ETV) and tenofovir disoproxil fumarate (TDF) groups with the absolute standardised mean difference below 0.1 indicating good balance among these parameters between the two groups.⁵ Although the above parameters were by no means exhaustive for the diagnosis of osteopenia or osteoporosis, most potential biases were mitigated for a fair and meaningful analysis. As a limitation of any retrospective study, some of the variables were not available, including the number of falls, degree of physical activity or blood selenium level, as suggested by Wu *et al.* We agree that the importance of these variables is not to be neglected, yet probably a prospective cohort, instead of the current study design, is needed to address all facets of the problem, and this may not be entirely feasible in such a large cohort and would delay a potential salient answer to a pressing research question given the time required for the events to occur.

In our study, most patients (96.1%) were treated with ETV and they were on the same therapy throughout the study period. Meanwhile in real-world clinical practice, due to renal and bone side effects, TDF users may switch to tenofovir alafenamide (TAF) but rarely would switch to ETV. Confined by the relatively small number of patients who were put on TAF with a short duration of treatment, the thoughtful suggestion of the analysis on switching therapy cannot be justifiably performed in our study. Nonetheless, there have been studies showing improved bone mineral density after switching TDF to TAF,^{6–8} as well as reduced risk of osteoporotic fracture using TAF instead of TDF in a treatment-naïve population.⁹ These serve as

the foundation for further evaluation of the actual incidence of clinical fracture after switching therapy.

In response to Dr. Wu *et al.*'s comment on the effect of age and sex on the study outcome, it is well-accepted that old age and female sex are risk factors for osteoporosis and fracture. In our study, otherwise, the effects of ageing and sex were well balanced after propensity score matching and these were equally impacting the ETV and TDF users. Indeed, the similar trend of cumulative incidence of incident fracture across those aged ≥ 65 years and ≥ 70 years in the sensitivity analyses strengthened our study conclusion. We appreciate that restricted mean survival time analysis would be an alternative method of tackling the non-proportional hazard of the treatment difference and would provide additional information along with the cumulative incidence functions.¹⁰ In our study, we utilised the time-dependent hazard ratio method to handle the non-proportional hazard, after detecting a time-varying treatment difference before and after month 24 based on the test of subdistribution hazards proportionality. The method aimed to provide a direct estimate of the treatment difference before and after month 24 instead of examining the overall difference in the area under the cumulative incidence functions as measured by restricted mean survival time.

After all, we echo Dr. Wu *et al.*'s insightful comments, respectfully comprehending the unanswered parts as a whole, and thus we believe further studies on this topic are essential.

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Conflict of interest

Jimmy Lai has served as an advisory committee member and a speaker for Gilead Sciences. Grace Wong has served as an advisory committee member for AstraZeneca, Gilead Sciences, GSK and Janssen, and as a speaker for Abbott, AbbVie, Asclelis, Bristol-Myers Squibb, Echosens, Gilead Sciences, Janssen, and Roche. She has also received a research grant from Gilead Sciences. Terry Yip has served as an advisory committee member and a speaker for Gilead Sciences.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

All authors were responsible for the study concept and design, interpretation of data, drafting and the critical revision of the manuscript for important intellectual content, and approved the final version of the article.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.05.009>.

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Author names in bold designate shared co-first authorship

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